

Aaditya Golash

In BC, willing to relocate

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Education

University of British Columbia

B.Sc. Computer Science, Minor in Management (Co-op)

BC

2021 – Present

- Relevant Coursework: Artificial Intelligence, Network Science, Analysis of Algorithms, Databases, Investments, Finance
- Professional Development: Adobe Experience Platform & Customer Journey Analytics Foundations, Google Analytics 4, Bloomberg Markets Concepts (BMC)

Professional Experience

Director

UBC Students' Union (UBCSUO)

May 2025 – Apr 2026

Kelowna, BC

- Represented 12,000+ UBCO students as an elected director while overseeing digital governance and funding decisions across a \$1.9M operating budget.
- Allocated \$370K+ in annual club and association funding to 150+ student organizations while chairing five committees on Policy, Funding, Campus Life, Campaigns, and Strategy.
- Contributed to SC14/SC11 policy updates for information systems and wireless network governance, improving digital policy outcomes for 70,000+ UBC students.
- Delivered engagement programs such as Orientation Expo, Cultural Festivals, Campus BBQs, and Community Meals, serving thousands of students and strengthening campus cohesion.

Student Support Analyst (Co-op)

UBC IT Services

May 2024 – Apr 2025

Kelowna, BC

- Resolved 110+ ServiceNow tickets daily across Wi-Fi, CWL, MFA, Microsoft 365, printing, hardware, and account systems for 12,000+ students and staff.
- Reduced repeat incidents and accelerated resolution by analyzing ticket patterns, documenting trends, and coordinating workflow fixes with UBC Vancouver IT.
- Rebuilt Student Support Knowledge Base article KB0019190 using HTML/CSS/TypeScript, improving discoverability and accessibility for campus-wide users.
- Supported Microsoft 365 migration and AI chatbot deployment, automating frequent requests and boosting support team capacity.

Projects

Tobii Pro Eye-Tracking Behavior Analytics

Research Pipeline Architect

Jan 2026 – Present

- Built a reproducible Python + Docker ETL pipeline for \$25K Tobii Pro Glasses 3 datasets, ingesting gaze, IMU, pupillometry, scene-camera, and event exports into synchronized behavioral metrics.
- Automated fixation, heatmap, Time to First Fixation, pupil-diameter, and confusion-event computation, turning raw sensor streams into research-ready HCI insights.
- Deployed a Flask/Dash upload service with interactive dashboards, making high-frequency usability data explorable for researcher review.
- Designed the pipeline around attention, ignored-content, confusion, and decision-point research questions so raw biometric streams could support interpretable usability evidence.

TSX Quantitative Portfolio Optimization

Data & Systems Lead

Jan 2026 – Present

- Built a TSX optimization engine processing 37 months of historical prices across 10 equities, converting raw market data into calibrated log returns for Modern Portfolio Theory.
- Modeled and validated an 11x11 variance-covariance matrix and solved Markowitz optimization with Solver-backed portfolio weights, prioritizing data integrity over output bias.
- Led the data-engine track for a 3-person team, producing risk-return frontier visualizations and quantitative insights used to assess portfolio efficiency and diversification tradeoffs.
- Created risk-return plots comparing individual securities against optimized allocations, translating matrix outputs into investment-analysis decisions.

SoccerNet Jersey Recognition Pipeline

Pipeline Architect

Jan 2026 – Present

- Improved jersey recognition accuracy from 87.12% to 88.52% across 1,211 tracklets by adding quality filtering, digit correction, and temporal consistency logic.
- Profiled a 5-stage computer vision workflow, separating ~900 images/sec preprocessing from ~78 images/sec PARSeq recognition to identify bottlenecks.
- Implemented Top-K frame filtering to rank legible frames by confidence, reducing noise from blur, occlusion, and poor camera angles in broadcast footage.
- Added quality filtering, digit correction, temporal consistency, and test-time augmentation; quality filtering produced the largest lift with +12 corrected tracklets.

TA Allocation & Management System

Backend Project Lead

May 2025 – Aug 2025

- Designed a Flask/MySQL allocation platform processing 250+ TA applications per term, replacing error-prone spreadsheet workflows and manual conflict checks.
- Implemented constraint-based assignment logic and role-based access control, cutting TA assignment processing time by 70% compared with spreadsheet workflows.
- Built analytics dashboards and administrative views across 36+ backend features, improving visibility into applications, assignments, and scheduling conflicts.
- Containerized deployment with Docker and GitHub workflows, supporting reproducible testing and cleaner handoff for future project maintainers.

Skills

Programming: Python, SQL, JavaScript/TypeScript, R, C/C++, HTML, CSS

Tools: Flask/Dash, Docker, GitHub Actions, REST APIs, OpenCV, Tableau, PowerBI

Data Engineering: ETL/ELT Pipelines, Sensor Data Processing, Database Design (MySQL, MongoDB)

Certifications: Bloomberg Markets Concepts, Adobe Experience Platform, Google Analytics 4, FOODSAFE Level 1